

BHARATHIAR UNIVERSITY: COIMBATORE – 641046
M.Phil. /Ph.D. - MICROBIOLOGY

PART I - SYLLABUS
(For the candidates admitted from the academic year 2018-19 onwards)

Paper – I: Research Methodology

Paper – II: Advances in Microbiology

Paper – III:

1. Recombinant DNA Technology
2. Immunotechnology
3. Bioprocess Technology
4. Environmental Microbiology
5. Vaccinology
6. Medical Microbiology
7. Industrial Microbiology

Paper – I: RESEARCH METHODOLOGY

Unit-I Introduction to Research Methodology

What is Research? Basic and applied research, Essential steps in research, Defining the research problem, Research/ Experimental design, Literature collection, Literature collection, Literature citation, Research report: components, Format of thesis and dissertation, Manuscript /Research article, Review monographs, Bibliography and Reference, Significance of research- Software for manuscript draft.

Unit-II Bio-statistical Methods

Measures of central tendency and dispersal; probability distributions (Binomial, Poisson and normal); sampling distribution; difference between parametric and non-parametric statistics; confidence interval; errors; levels of significance; regression and correlation; t-test; analysis of variance; X^2 test; basic introduction to Multivariate statistics, etc.,

Unit-III Applied Bio-statistics

R Language and its simple applications – Computation of Probabilities and cumulative Probabilities using Binomial and Poisson models. Evaluation area and ordinate under normal distribution using R Software, SPSS.

Unit-IV Research Ethics & IPR

Perspective of Ethics, Personal vs professional ethics: Moral Reasoning – Ethical theories, Deontological, Utilitarianism – Ethical leadership (integrity and ingenuity) - framework for ethical decision making- Michael Macdonald model & Storch model, Plagiarism software

Introduction to intellectual property and intellectual property rights Types, patents, copy rights, trade marks, design rights, geographical indications – importance of IPR – patentable and non patentables – patenting life – legal protection of biotechnological inventions – world intellectual property rights organization (WIPO)

Unit-V Biosafety guidelines

Introduction to biosafety Biosafety issues in biotechnology – risk assessment and risk management – safety protocols: risk groups – biosafety levels – biosafety guidelines and regulations (National and International) – operation of biosafety guidelines and regulations – types of biosafety containment; Depository regulation- National & International centers; Biological databases.

References:

- C.R. Kothari, IInd edition (2004) Research methodology, Methods and techniques, New Age International (P) Ltd, Publishers, New Delhi.
- Jenod H. Zar (1999) Biostatistical analysis by, Prentice Hall International, Inc. Press, London.

- Everitt, B. S., and Hothorn, T. (2010). A Handbook of Statistical Analyses Using R, Second Edition, Chapman and Hall, CRC Press.
- Sabine landau and Brain S.Everitt (2004). A Handbook of Statistical Analyses Using SPSS. Chapman and Hall, CRC Press
- Quick, J. M. (2010). Statistical Analysis with R, Packt Publishing Ltd., UK.
- Principles of cloning, 2nd E6dition, Jose Cibelli, Robert Lanza , Keith H.S. Campbell, , Michael D. West, 2013
- Ethics in Engineering 4th Edition by Martin, Mike W.; Schinzinger, Tata McGraw Hill Science, 2005
- Ethical Issues in Biotechnology, Richard Sherlock, John D. Morrey, Jan 1 2002, Rowman & Littlefield.
- <http://books.cambridge.org/0521384737.html>

Paper – II: ADVANCES IN MICROBIOLOGY

Unit-1

Microscopic Techniques, SEM, TEM, Confocal and Fluorescence- Imaging –Working Principles and its applications.

Unit-II

Biophysical Techniques: UV-Visible, Fluorescence, Circular NMR, FT-IR, X-ray-surface plasma resonance methods- working principle, applications and data analysis.

Unit-III

Advanced diagnostic procedures in Microbiology – Culture confirmation techniques; Direct detection probes; Diagnostic sequencing; Molecular typing methods; Pulse Field Gel Electrophoresis; PCR based typing methods; Genotyping of bacteria by using VNTR. Agglutination and Precipitation – EIA, ELISA, Immunofluorescence, RIA, Chemiluminescence, Blotting techniques (Western, Southern, Northern), Flowcytometric assays. Automation in diagnostic microbiology. Bactec, Vitek, MALDI-TOF.

Unit-IV

Metabolic Engineering, Targeted gene replacement, Gene Knockout, Gene silencing techniques (Antisense RNA technology & RNAi) Genome editing (CRISPR/Cas9) Protein sequencing. Metagenomics: Introduction; Pure culture and in consortium; Cultivable and Non-cultivable microbial analysis, Differential expression analysis (DEA); Microarrays & Metagenome sequencing; Next-generation sequencing approaches to metagenomics.

Unit-V

Downstream Processing- Removal of microbial cells and solid matter, foam separation, precipitation, filtration, centrifugation, cell disruptions, liquid-liquid extraction, chromatography, membrane process, drying and crystallization. Quality analysis and product formulation - Product stabilization and shelf life analysis and quality control. Fermentation economics of citric

acid – process economics and project economics. Separation Techniques – Chromatography Techniques; TLC, HPTLC, HPLC, FPLC, UPLC, LC-MSMS, GC-MS.

REFERENCES

- J. Sambrook, D.W Russel, Molecular cloning: A laboratory Manual, Vol.1,2,3 Cold spring Harbour Laboratory Press,2001
- Pranavkumar, Fundamentals and Techniques of Biophysics and Molecular Biology-2016
- Albert L. Lehninger, David L. Nelson, and Michael M. Cox, Principles of Biochemistry
- Christopher J. Woolverton, Joanne Willey, and Linda Sherwood. Prescott's Microbiology.
- Kevin Robards, P. E. Jackson, Paul A. Haddad. Principles and Practice of Modern Chromatographic Methods. 2016
- Principles of gene manipulation and genomics S.B.primrose and R.M Twyman Seventh Edition.
- Sharma.BK.Instrumental methods of chemical analysis.
- Upadhyay, Upadhyay and Nath, Biophysical chemistry.
- KhandpurR.S. Handbook of biomedical instrumentation ,Tata Mc GrawHill.

PAPER - III**1. RECOMBINANT DNA TECHNOLOGY****Unit-I**

DNA replication, enzymes, accessory proteins and mechanism of prokaryotic & eukaryotic DNA replication. Transcription: Organization of transcription unit, mechanism of transcription & its regulation in prokaryotes & eukaryotes (general and specific transcription factor regulatory elements, Translation, genetic code, prokaryotic & eukaryotic translation, regulation of translation post translational modification of proteins.

Unit –II

Principles of genetic engineering: Cloning strategies: RE digestion, cohesive and blunt end ligation, use of linkers and adapters in generation of recombinant DNA, significance of dephosphorylation. Construction of vectors–pET types Strategies for introducing recombinant molecules into host. Transformation, particle bombardment, conjugation

Unit- III

Construction of genomic libraries and cDNA libraries – identification of specific genes selection for an expressed phenotype-Hybridization-Techniques with nucleic acid probes-Immunological Techniques.: Molecular characterization of cloned genes– Restriction analysis-DNA Sequencing- Analysis of transcripts. Site specific mutagenesis. Overexpression of genes in *E.coli*.

Unit –IV

Genetic engineering of bacteria – methods for the genetic manipulation of Bacilli- gene expression. Genetic engineering of Streptomyces – methods of gene manipulation – gene expression –use of Streptomyces as a host for excretion of heterologous products.

Unit –V

Genetic engineering of yeast. Molecular techniques for gene manipulation of *Saccharomyces cerevisiae*. Molecular transformation – selection markers – vectors – Expression of Heterologous proteins. Genetic Engineering of Filamentous fungi for industry – antibiotics, Proteins production – fungal enzymes. Mammalian proteins

References:

- Genetic Engineering of microorganisms. Ed. By Alfred publishers. 1993. VCH. Weinheim FRG and New York USA.
- Old R.W and Primrose S.B .1995. Principles of gene manipulation-An introduction to genetic engineering. 5th edition. Blackwell scientific publications. London.
- Winnaker E.C. 1987. From genes to clones. Introduction to gene technology. VCH. Weinheim.

2. IMMUNOTECHNOLOGY

Unit-I:

Immunogens- Antigens - properties and types. Haptens. Epitopes-properties of B and T cell epitopes, Tumor- specific antigens. Preparation of Bacterial, fungal and viral antigens for raising antibodies. Handling and immunization of laboratory animals (Mice, Rabbit, Chicken), adjuvants and their mode of action.

Unit- II:

Immunoglobulins- structure, properties, types and subtypes. Production of polyclonal and monoclonal antibodies. Therapeutic application of monoclonal antibodies for human disease. The role of monoclonal antibodies in the advancement of immuno assay technology. Immunoaffinity purification of antibodies. Modification of antibodies by chemical methods.

Unit-III:

Immunohematology - ABO and Rh factor significance, practical application of immuno methodology in blood transfusions, Erythroblastosis foetalis. Transplantation: Terminology, Auto graft, Isograft, Allograft, Xenograft, organ transplantations in humans, HLA typing methods, Graft rejection- Graft disease (GVHD).

Unit-IV:

Prevention and therapy of immunologic diseases- vaccines- live attenuated- killed, subunit, antitoxins, Antivenom, vectored vaccines, nucleic acid (DNA) vaccines. Currently licensed vaccines- Recombinant delivery systems for future vaccines- New approaches for better vaccines. Practices of immunization- therapeutic principles- new approaches to immunization- mucosal vaccine, maternal immunization. Immunotherapy of allergic disease – Interleukins and Interferons. Treatment of malignancies- Experimental immunotherapy of auto immune disease- Recent research and development of AIDS vaccines.

Unit- V:

Immunodiagnosis- methods for immunoglobulin determination – Quantitative and qualitative antigen and antibody reactions. Agglutination, precipitation Immunofluorescence, Western blot analysis, Immuno Electrophoresis, SDS-PAGE, ELISA-Principle and Applications, Radio Immuno Assay (RIA) Principles and Applications, Non Isotopic Methods of Detection of Antigens - Enhanced Chemiluminescence Assay.

References:

- Dm. Weir., 1986. Handbook of Experimental Immunology vol – I-IV, Black well publication.
- Robert R. Rich. 2001. Clinical Immunology Principles and practice. Mosby publication. Charles A. Janeway., Paul Travers, Mark Walport and Mark Sclomcik. 2001.
- Immunobiology- The Immune system in Health and Disease. Churchill Livingstone publication. New York.
- Kuby, Richard A. Goldsby., 2000. Immunology. 4th edition.
- Robert M. Coleman., 1992. Fundamental Immunology. 2nd edition., Wm. C. Brown Publishers.

3. BIOPROCESS TECHNOLOGY

Unit-I:

Isolation of cultures - screening of new products from microorganisms – inoculum development – scale up of microbial, mammalian and plant cell processes – selection of scaleup criteria – strain improvement – screening and selection for strain improvement – protoplast fusion and mutagenesis.

Unit –II:

Bioreactions – bioreactors in bioprocessing of cells – enzyme bioreactors – biosensors in bioprocessing – shake and solid state fermentations – archaeal bacterial bioprocessing by cell synchronization – recent advances in yeast bioprocessing – multiinteracting microbial bioprocessing – variables in bioprocessing.

Unit – III:

Basic concepts in medium design – design procedure growth limiting nutrient in designed medium – cell growth and product formation – immobilization and cell culture – cell immobilization – enzyme immobilization – mammalian cell culture. Bio instrumentation and Computer control of fermentation processes.

Unit – IV:

Bioproducts - modern estimation and bioassay – sterilization – medium sterilization – batch and continuous sterilization – sterilization of fermentor – feeds, liquid wastes and filter sterilization –
Bioprocess of treatment of liquid wastes and water.

Unit –V:

Bioproducts recover – conventional recovery methods. Cell disruption, foam separation, centrifugation, cell disruption – liquid - liquid extraction, counter - current distribution – chromatography. Newer methods – cross filtration and cassette separation, dispersive separations.

References:

- Mukhopadhyay, S.N., 2001 Process–biotechnology fundamentals. Viva books (P) Ltd.
- Arnold. L . Demain and Nadine A.Soloman . 1986 . Manual of Industrial and Biotechnology. American society for Microbiology, Wasington DC.Stanbury, P.F., A.Whitaker and S.J.Hall. 1995. Principles of fermentation technology-Second Edition-Elsevier Publications.

4. ENVIRONMENTAL MICROBIOLOGY

Unit I: Bio remediation

Reclamation of polluted environments (soil, water, effluents) with microbes, Microbial interaction with xenobiotics, Sewage and water treatment, Bio waste management, Treatment of toxic metals with microbes, Case studies of bio remediation strategies.

Unit II: Soil Microbiology

The Rhizosphere- Mycorrhizal symbiosis (Ecto , Endo), Nitrogen fixers (Azospirillum, Azotobacter, Rhizobium, Frankia (Actinomycete), Nitrogen cycle, Carbon cycle, Role of Microbes in global warming and Carbon sequestration, Bio control of soil pathogens.

Unit III: Plant – Microbe interactions

Tripartite relationship of legume, Rhizobium and Mycorrhizas, Beneficial microbes for crop improvement, Plant growth promoting rhizobacterias in Agriculture and Forestry, Cyanobacteria as bio fertilizers, Azolla and Cyanobacterial symbiosis, *Pirifomspora indica* and their role in plant improvement.

Unit IV: Aquatic Microbiology

Microbes in aquatic habitats (pond, lake, streams, estuaries) and their importance, Mass multiplication techniques of beneficial blue green alage, Methanogens, Role of microbes in Mangrove eco systems, Microbial pathogens of Fish and control.

Unit V: Microbial products for Agriculture

Bio formulations of beneficial microbes/bio fertilizers (liquid and solid), Fermentation and storage techniques for mass multiplication of microbes, Application of bio fertilizers in seeds, seedlings for crop productivity, Microbial products for purification of water.

References

- Text book of Environmental Microbiology (2008) by Pratiba K. Mohabatra
- Environmental Microbiology for Aquatic and Waste systems (2011) by Nduleka Okafar
- Hand book of Microbial bioferiliizers (2006) by M.K. Rai, Int. Book Distributors Co, New Delhi
- Agricultural Microbiology (1992) by G. Rangasamy and D J Bagyaraj Subbarao N.S. (1982) Advances in Agricultural Microbiology

5. VACCINOLOGY

UNIT-I:

Immune system –recognition of nonself and self ; Humoral Immunity –immunoglobulins, basic structure, classes and subclasses; Cellular Immunity - lymphocytes, lymphokines, cytokines and interones; Antigen Recognition – membrane receptors for antigens, MHC classification and functions, superantigen.

UNIT-II:

Recombinant vaccines; polynucleotide as vaccines; vector vaccines; naked DNA vaccines; biosynthetic and chemically synthesized vaccines; subunit vaccine; antiidiotype vaccines; fusion vaccines; mixed particle vaccines; human mucosal vaccines; Combination vaccines; Edible vaccines produced in transgenic plants and microencapsulation.

UNIT-III:

EPI vaccines–production and testing of tetanus toxoid, diphtheria toxoid, pertussis vaccine, BCG vaccines; preparation of Hepatitis B vaccine and tissue culture derived rabies vaccine and AIDS vaccine.

UNIT-IV:

Adjuvants –classification and properties; carriers–types and functions–vehicles –types, functions and mode of action ; biodegradable polymers - micropheres, liposomes and ISCOM ; immunostimulators (IS) –classification and mechanism of action of IS.

UNIT-V:

Germ free , Axenic, Monoxenic, dixenic, Gnotobiotic, conventional, xyloxic, SPF; isolators, Gnotobiotic technique, gene knockout mice; transgenic and SCID mouse; Athymic nude mouse; animal models of human diseases and techniques of experimentations.

References:

- Gregory G. et al; 1995 Vaccines; new generation immunological adjuvants. Series A: Life Sciences; Volume: 282.
- Heing K. and T. L. Philip: 1986 Vaccines: New concep
- ts and developments –proceedings of the 10th international convocation on immunology; July 14 –17. buffalo, New York. Longman scientific and technical 1998; New York.
- Gregory G., C.A.Anthony and P.George; 1990. Vaccines: recent trends and progress . Series A : Life sciences: vol; 215 plenum press, New York

6. MEDICAL MICROBIOLOGY**Unit – I:**

Introduction of medical microbiology – Infection Diseases process – diagnosis – process of sample collection, transport and examination of the specimens. Antibigram-Drug resistance-format of reporting the culture and serology results.

Unit – II:

Bacteriology: gram-positive – morphology culture characteristics, pathogenicity and laboratory diagnosis of *Staphylococcus aureus* , *Streptococcus pyogenes*, *Pneumococcus*, *Bacillus anthracis*, *Corynebacterium diphtheriae*, *Clostridium welchii*, *Clostridium tetani*, *Clostridium botulinum*, *Mycobacterium tuberculosis*, *M. leprae*, *Spirochaetes- Treponema pallidum*, and *Leptospira lacterohaemorrhagiae*.

Unit –III:

Bacteriology gram negative organisms: morphology culture characteristics, pathogenicity and laboratory diagnosis of *E.coli*, *Klebsiella* spp, *Enterococcus* sp, *Salmonella typhi*, *Shigella dysenteriae*, *Pseudomonas* sp, *Vibrio cholerae*, *Acinetobacter baumannii*, *Aeromonas hydrophila*, *Bordetella pertussis*, *Yersinia pestis*, and *Neisseria* spp.

Unit – IV:

Virology: Basic concept of virology – general properties of human viruses, Approaches to viral diagnosis – serological and molecular techniques. Pathogenicity and laboratory diagnosis of viral infections- Hepatitis, polio, Rabies, Influenza, Measles, Mumps, Rubella, Dengue virus, HIV.

Unit – V:

Mycology: General properties and approaches to laboratory diagnosis. Mycosis - superficial, Subcutaneous and Systemic infections- Cryptococcosis, Madura mycosis, Histoplasmosis, *Candida albicans*.

Parasitology: pathogenicity and laboratory diagnosis of *Entamoeba histolytica*, *Taenia solium*, *Plasmodium vivax*, *Wuchereria bancrofti* and *Enterobius vermicularis*. *Trichomonas vaginalis*.

References:

- Ananthanarayan, R. and Paniker, C. (1980). *Textbook of microbiology*. 1st ed. Orient Longman.
- Krishnadhan Chatterjee, Parasitology, (2009). Protozoology and Helminthology in Relation to clinical Medicine. CBS Publishers & Distributors.
- Shimeld, Lisa Anne & Rodgers, Anne T (1999). *Essentials of diagnostic microbiology*. DelmarPublishers, Albany, NY.
- James H Jorgensen; Michael A Pfaller; Karen C Carroll, (2015). *Manual of clinical Microbiology*, Washington, DC : ASM Press
- Textbook of Medical Parasitology : *Protozoology & Helminthology*, (2013). 4th Edition by Subhash Chandra Parija, Ahuja Publishing House.
- Geo. F. Brooks, Karen C. Carroll, Janet S. Butel, Stephen A. Morse. (2007). *Jawetz, Melnick & Adelberg's Medical Microbiology*, 24th edition, McGraw Hill Professional.
- James H Jorgensen; Michael A Pfaller; Karen C Carroll; (2015) *American Society for Microbiology, Manual of Clinical Microbiology*, Washington, DC : ASM Press.

7. INDUSTRIAL MICROBIOLOGY

UNIT I -

History and chronological development of industrial microbiology. Industrially important strains – Isolation and preservation. Inoculum development for various fermentation process. Strain development – mutation, recombinant DNA technology and plasmid fusion.

UNIT – II

Fermenter – design and types - Fermentation types: solid, submerged, Batch and Fed batch and Continuous fermentation. (Synchronous, diauxic, batch) Downstream processing - Extra and intracellular process – Column chromatography, and gel filtration.

UNIT – III

Production of secondary metabolites: Antibiotics, microbial enzyme (amylase, alkaline protease and xylanases) – production, purification and immobilisation. Single cell protein – baker's yeast and Spirulina - cultivation of mushroom. Production of bacteriocins and antibiotics using probiotic bacteria. Biotransformation of heavy metals and xenobiotics - Petroleum biodegradation: reductive and aerobic dechlorination. A brief account of biodegradable plastics and super bug.

UNIT – IV

Clinical uses of antimicrobial drugs, Microbial spoilage and preservation of pharmaceutical products, Sterilization of pharmaceutical products, Applications of microorganism in the pharmaceutical sciences.

UNIT – V

Role of precursors and steering agents in production of antibiotics, vitamins. Antiseptics - disinfectants their standardization and Quality control of Pharmaceutical products – Indictable, IV fluids and pyrogen testing.

References:

- Microbiology by Prescott, Prescott, Harley and Klein's 5th edition McGraw-Hill publishers
- Fermentation Technology by Stanbury & Whitaker, 2nd edition, Butterworth-Heinemann Limited.
- Text book of Industrial Microbiology, 2nd edition by Wulf Crueger and Anneliese Crueger, Sinauer Associates Inc
- Industrial Microbiology, 1983, 4th Edition, Prescott and Dunn's, Gerald Reed, AVI Publishing Company Inc. Connecticut
- Environmental Engineering, Kiely, G (1998) Irwin/McGraw Hill International, U.K.
- Hazardous Waste Management, II Ed, LaGrega, M.D., Buckingham, P.L., and Evans, J.C (2001) McGraw Hill Inc.
- Microbial Degradation of Xenobiotics and Recalcitrant Compounds, Leisinger, T, Cook, A.M., Hutter, R and Nuesch, J (1981) Academic Press, London.